

Predictions for Formula 1 Races

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Background

What we wanted to do:

- Predict 'excitement' of a race

Key Stakeholders:

- New and Existing Fans
- FIA



Data

Source: [f1db](#)

What data was available

- Qualification times
- Equipment manufacturers
- Driver information
- Course information

What we chose

- Gap time
- Circuit type
- Starting Grid



Target and Features

- Target: Average Local Position Change

$$\frac{1}{N(N-1)} \sum_{i=1}^N \left| \sum_{j \neq i} (i - j) - (\sigma^{-1}(i) - \sigma^{-1}(j)) \right|$$

σ = finishing positions

15234

13524

- Features

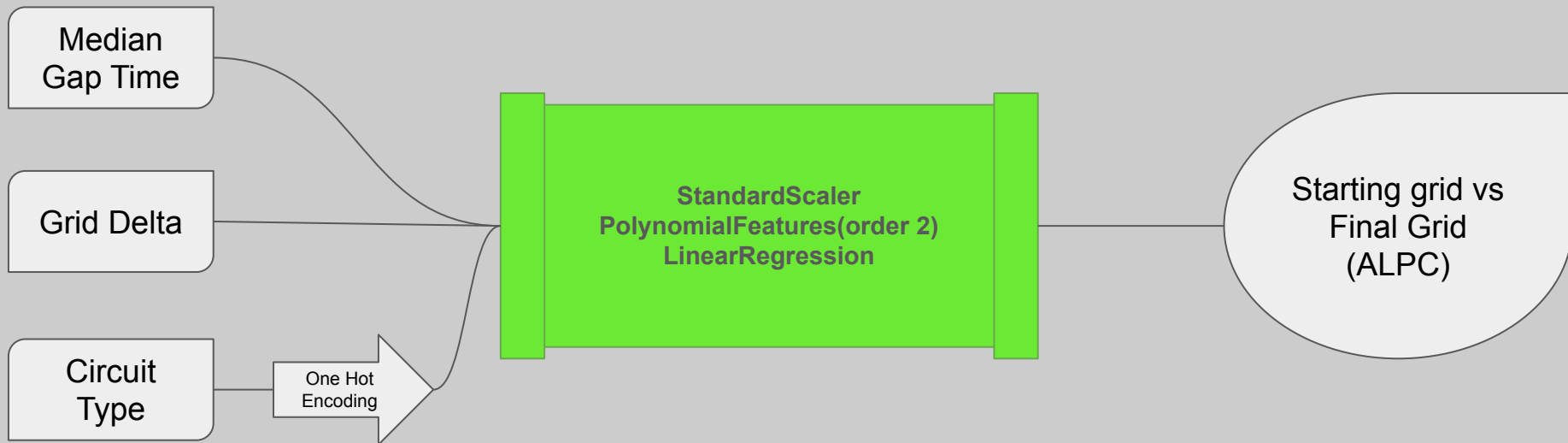
- Absolute Grid Delta: Average absolute difference between drivers' pace ranking based on best lap time from FP/Qualifying and their starting position.
- Median Gap Time: Derived from calculating gaps in fastest qualification times between each driver and the next fastest driver

$$\frac{1}{N} \sum_{i=1}^N |Grid[i] - paceRank[i]|$$

Median Gap Time = $Median(FQT[2] - FQT[1], FQT[3] - FQT[2], \dots, FQT[N] - FQT[N - 1])$

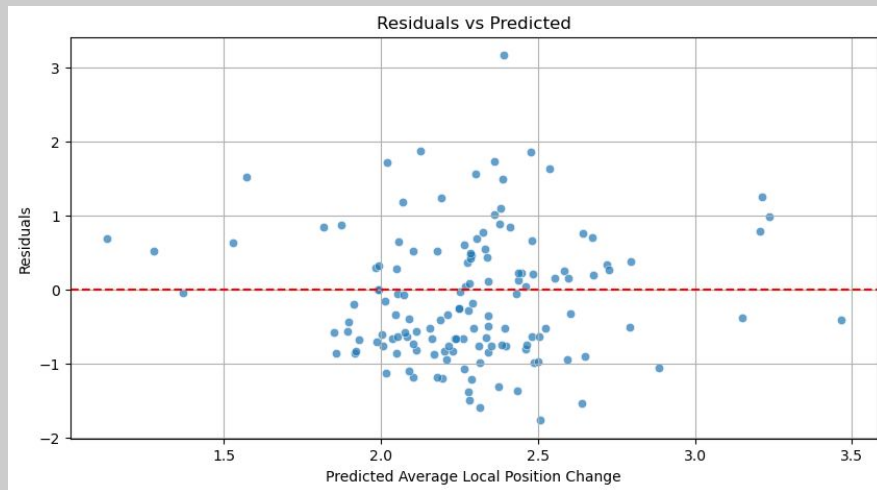
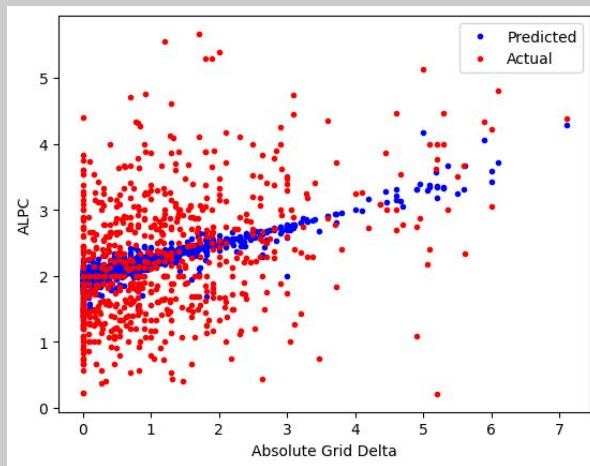
FQT = indexed list of Fastest Qual Time per race

Model



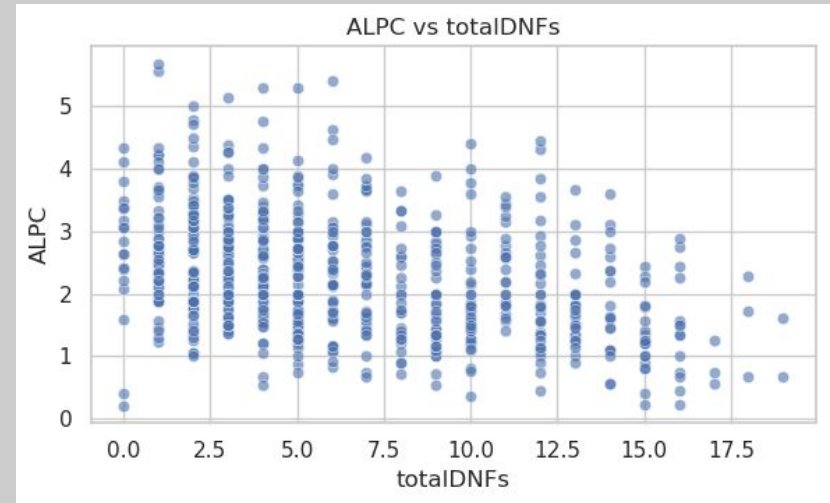
Model Outcomes

- Our model is able to predict ALPC with mild success
 - Achieves an r-squared of approx. 0.13
- The most predictive feature was absolute grid delta, with median gap time and circuit type providing slight enhancements



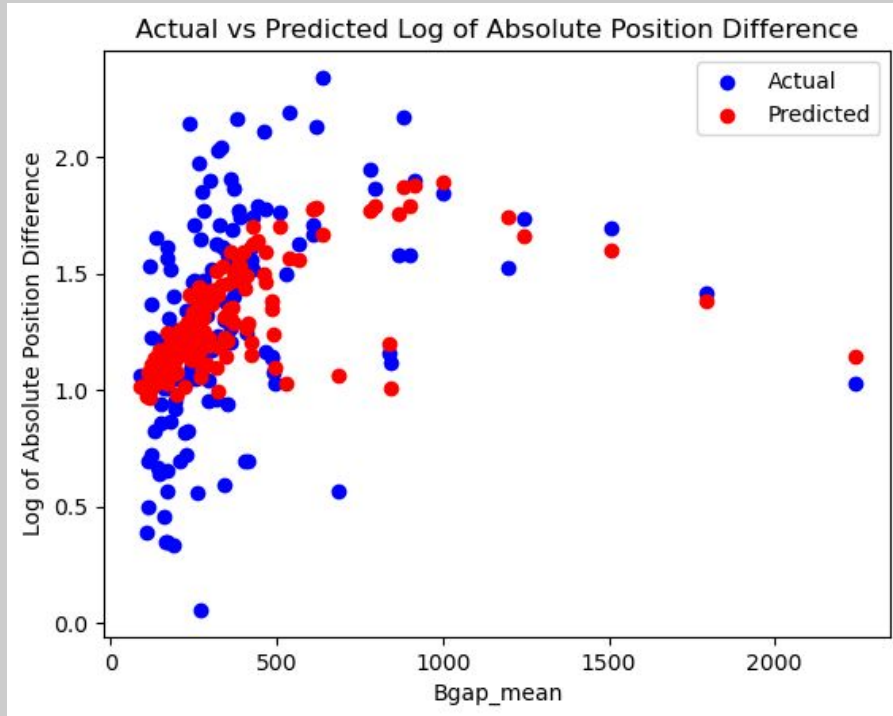
Explaining Model Performance

- It appears that significant variance is caused by “random” occurrences during the race, such as DNFs.
 - Refitting the model with DNFs excluded yields an improved r-squared score of 0.20
 - Events like these cannot be known ahead of time, providing a challenge for predictions
 - Wildlife collisions, weather events, human error, etc. are other such unknown events causing variance
- Further, DNFs appear to be correlated with ALPC



Additional Observation: Interesting gap trend

Log of Mean gap between qualifying times vs. Average net position changes



R-squared: 0.393

RMSE: 1.43

Possible explanation:

- Small gap = less room for maneuvers
- “Just enough” gap = close enough to pass and enough room
- Too much gap = can’t pass if the car is too far ahead.

Conclusion & Limitations

01 — Absolute Grid
Delta proved to
be the most
useful feature

02 — Current
approach has
notable
limitations



03 — Overtakes
happen when
the fastest car
starts from the
back

04 — Random events
influence race
outcome significantly
(e.g. weather,
crashes, or
mechanical failures)

Future Work

- Include historical features (like circuit-specific data)
- Try to predict DNFs in a race, and use it as a feature for our “excitement” model
- May use machine learning to process the data and find features that might influence the outcomes

